

Cooperative Self-Scheduling Routing Approach Based on Energy Efficient Optimal Link Stability Routing Allocation for Improving QoS-WSN

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Abstract—The development of Wireless Sensor Networks (WSN) is broad research in recent communication approaches. Due to the increasing demand for service, the quality is insufficient to meet the requirements to take energy efficient network. This leads to high energy losses, traffic, improper scheduling, and poor network performance. Selecting a higher traffic route introduces higher packet drop and reduces the throughput performance. Analyzing the traffic in the existing process is complex and doesn't provide Quality of Service (QoS). So data analysis is essential based on reinforcement learning with an intelligent machine learning approach. To resolve this problem, we propose a Cooperative self-scheduling routing approach (CS2RA) based on energy Efficient Optimal Link stability routing allocation (EOLSR) designed to improve the quality of service in Wireless sensor networks. The dynamic network topology medium is constructed with Neighbor Stable Link Connectivity Based Routing. With the Probability support, the traffic density approximation rate (PTDAR) is estimated by improving energy stabilization. By intending an optimistic Q-learning network model based on the energy-consuming parameters using an adaptive Petal spider ant colony cluster (PSACC) is constructed for link stability for route optimization. Then Cooperative stabilized proactive Energy-aware cluster routing protocol (CSPECR) was intended to make self-balanced congestion routing with the support of the Lookup energy constraint duty cycle (LECDC) to improve the lifetime of the network.

Index Terms—Link stability, cooperative communication, cluster routing, Lifetime WSN, self-scheduling routing.

I. INTRODUCTION

In recent research, Q-learning algorithms have been optimistic in reinforcement learning models to analyze the network data based on the behavioral approach to enhance network routing. Wireless communication is a developing technology in recent day empowerment for sensing, monitoring, and transferring data in remote areas based on a heterogeneous model. By handling the data transfer, it is difficult to manage the energy

level to route the data [1]. Most Q-learning network models take the feature dimensions at a high level by considering more non-relative features. Because energy consumption leads to packet loss, delay tolerance, and cause latency failures to break the network's lifetime.

To achieve the data broadcast, energy-efficient routing is developed in WSN [2]. The source node identifies the route and performs cooperative transmission with the support of other nodes. The source node with data to be communicated to the destination node determines the routes available to reach the destination. According to the routes, open Cooperative self-scheduling routing protocol is developed [3]. A single route according to the routing strategy is selected to perform data transmission.

To propose a Cooperative Self-Scheduling Routing Approach (CS2RA) optimized based on self-balanced cooperative communication in the wireless sensor network. Initially, the neighbor discovery node level depletion is estimated and the Node response behavior rate finds the active transmission level which considers the energy delay transmission drop ratio and latency to estimate the absolute mean weight for considering the active nodes. By the intensive Q-learning estimation adaptive Petal spider ant colony cluster was constructed for link stability for route optimization. Then Cooperative stabilized proactive Energy-aware cluster routing protocol was intended to make self-balanced congestion routing with the support of the Lookup energy constraint duty cycle (LECDC) to improve the network's lifetime.

This proposed system performs energy-efficient data transfer to improve the delay tolerance in latency level to achieve performance in higher throughput to make better lifetime communication with the support of Hierarchical clustering-task scheduling [4]. This leads to high tip power dissipation

and short tip lifetime. It also has a method of sending/receiving data by considering the traffic of nodes and routes along the cooperative route and selecting the route with the least traffic [5]. It improves performance efficiency and requests root power from one or more nodes. Compared to common transmission models this proposed system improves network lifetime and QoS performance and consequently generates higher throughput.

The proposed system evaluates the routing traffic energy level and cooperative multipath sharing followed by packet density estimation to assess the data transmission size for each window period. To evaluate the energy consumption in the overall network, the proposed performance evaluation method using various QoS matrices and comparing it to the existing method provides better performance.

II. RELATED WORK

This section presents a detailed review of the routing methods and discusses the comparative study of various methods.

The broadcast nature of radio communication channels equips devices to communicate via every device within its aligned broadcasting region. Whenever a device acquires data, no other device neighboring isolates from the transmitter and must transfer the data. The device approaches a common intermediate whenever its communication cannot trouble any fixed session [6].

Unlike proactive routing protocols, reactive routing protocols do not initiate the route discovery process until each node requires a route, so reinforcement learning covers the routing capability implemented in VANET progress described in [7]. By applying reinforcement learning, the neighbor routing approach is mostly covered by distance routing protocol with classified routing approach described to use service dependency [8] and failure data collected in the network through distributed runtime monitoring of service interactions and failure manifestations. One of the main contributions of DSR is the absence of particular beacon messages (or hello packets) to receive local state information.

An energy-based multipath routing algorithm using Particle Swarm Optimization (PSO) is presented, which considers reliability measures such as energy cost and traffic for routing between sources and destinations [9]. This protocol uses the source routing principle and stores the entire route from the start to the goal during the route construction phase. A trust-based multipath routing algorithm called Ad Hoc On-Demand Trusted-Path Distance Vector (AOTDV) is proposed and presented for MANET. The protocol can find multiple loop-free routes in a single route search [10].

A hybrid routing algorithm for delay-tolerant networks is presented considering the life traces of dynamic topologies. Compare method performance with DSR and SBOR. An efficient power-aware multipath routing algorithm is presented which considers power, the distance of transmission, and hop count in the route selection [11]. An efficient fitness function-based energy-efficient multipath routing algorithm is presented which clusters the nodes.

Trust-based routing is enforced using a fuzzy rule analysis scheme. It maintains a set of rules for different types of populations and scales confidence values according to feature values and available practices [12]. The DPSRB approach improves efficiency. A static energy-based routing scheme is presented considering node dynamics and stability between other nodes. Accordingly, transportation and energy are used for path selection [13].

An efficient load-balancing algorithm is presented for the scheduling approach where data forwarding is performed through the intermediate nodes. It considers the traffic and latency in the network. Load balancing in multipath routing is presented which considers energy consumption as the key to selecting the route. The route selection is performed according to network failure, energy depletion, and congestion [14].

A location-based routing scheme is presented which performs security enforcement by partitioning the network using the Optimal Tug of War partition (OTW) algorithm and measures the signal strength to find an optimal secure route. A triple factor-based secure routing scheme is presented in [15]. The triple factor includes reputation among others in different ways to compute the trust value which identifies the secure route according to the earlier behavior of different nodes. A trust-based secure routing scheme is presented in [16] which computes trust factors according to recent trust, historical trust, and direct/indirect trust.

An ant colony optimization-based cluster base routing scheme is presented which identifies the route according to the trust values. An AODV-based secure routing scheme is presented in [17]. Similarly, a trust-based routing scheme is presented which measures the trust value according to the direct and indirect interactions. Based on that, an optimal route is selected to perform routing [18].

The efficient multipath routing algorithm over AODV is presented which identifies the list of nodes in the network and performs route discovery to perform data transmission [19]. The author presents an analysis on the performance of different algorithms namely Petal Ant Routing (PAR) and Dynamic Source Routing (DSR) [20], [21]. The behavior of this kind of Flooding Attacks Prevention Routing Protocol (FAPRP) [22], [23].

Secure Token Based Multipath Routing Protocol is discussed which uses hybrid clustering and path selection algorithm [24]. The performance of the algorithms is measured and compared with the performance of other methods. In [25], an efficient Fuzzy Controllers Based Multipath Routing Algorithm is discussed which enables the establishment of multiple paths between different sources and destinations.

Cooperative routes in the network are assumed to be linear and strip-shaped where each hop or a level contains multiple nodes and the distance between the nodes is kept constant. We assume all nodes know their relative location in advance. By maintaining linearity, the aim is to generate an energy-efficient route by selecting any active node in each level. The proposed system performs energy-efficient data transfer to improve the delay tolerance in latency level to achieve performance in

higher throughput to make better lifetime communication.

III. PROPOSED METHODOLOGY

Towards the development of QoS, we propose a Cooperative self-scheduling routing approach based on Energy Efficient Optimal Link stability routing. This method uses neighbor information and links between neighbors to predict link quality. It maintains state between nodes by monitoring the state of neighboring nodes. Using this information, the method performs analysis at various stages to identify stable connections for data transfer. This approach improves routing performance and achieves higher values for various QoS parameters.

We propose a Cooperative Self-Scheduling Routing Approach (CS2RA) based on Energy Efficient Optimal Link Stability Routing Allocation (EOLSRA) to improve the quality of service in Wireless Sensor Networks (WSN). The selection of higher traffic routes introduces higher packet drop and reduces throughput performance. It is difficult to analyze traffic in the existing process, and it does not provide the required Quality of Service (QoS). We present Neighbor Stable Link Connectivity Based Routing. The Probability Traffic Density Approximation Rate (PTDAR) is estimated to improve energy stabilization. Based on this estimation, an adaptive Petal Spider Ant Colony Cluster (PSACC) was constructed for link stability and route optimization. Then, the Cooperative Stabilized Proactive Energy-Aware Cluster Routing Protocol (CSPECR) was developed to achieve self-balanced congestion routing with the support of Lookup Energy Constraint Duty Cycle (LECDC) to improve the network's lifetime.

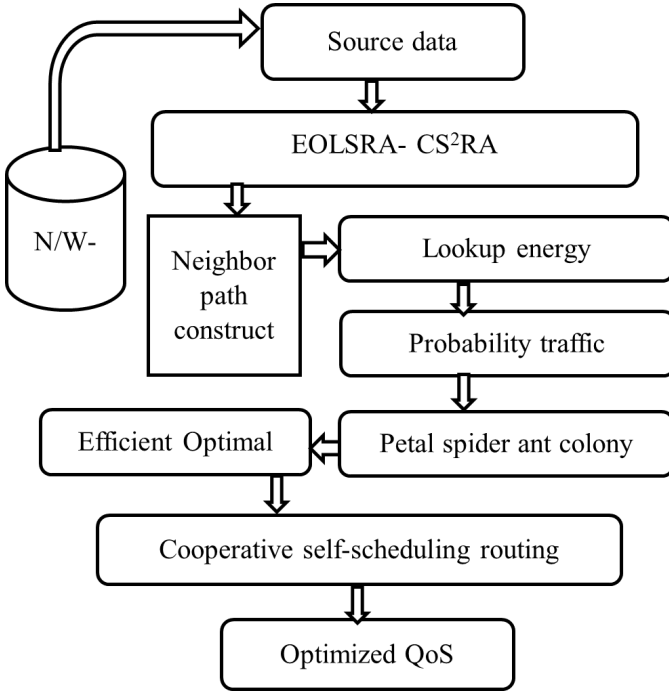


Fig. 1. Proposed architecture EOLSRA-CS2RA

This introduces cooperative transmission in the network to perform data communication. Figure 1 shows the proposed

architecture EOLSRA-CS2RA. A source node finds available routes and selects a route to perform data transmission. Data packets are sent through intermediate nodes to complete the data transfer. However, path selection plays an important role in achieving high operational efficiency and other constraints.

A. Lookup Energy Constraint Duty Cycle (LECDC)

In this stage, the energy level constraints are monitored in routing through cluster heads. The maximum energy nodes are avoided, and the sensor nodes are assigned as reactive nodes to transmit data. However, the active nodes' energy consumption leads to significant depletion, affecting network lifetime. The aim is to manage the active nodes (n) in the cluster head response (K). The network lifetime is increased by each duty cycle through lookup energy consumption in cooperative communication routing in a multicast cluster propagation.

By defining R as the route in Lookup transmission, which contains all the energy constraint nodes in the multicast network $T(s, D)$, where $R \rightarrow$ source 's' and destination 'D' create a cluster group set $C_G \rightarrow D \subseteq \{R - \{s\}\}$.

The Relation Lookup Transmission is given by:

$$R_T(s, D) \rightarrow D \subseteq \{R - \{s\}\} \quad (1)$$

Considering the minimum energy at delay constraints,

$$\text{Energy}(p(s, d)) = \min\{R_T(e), e \in p(s, d)\} \quad (2)$$

$$\text{Traffic}_{psd} = \min_{RT} \sum_{e \in psd} \text{delay}(e) + \sum_{e \in psd} \text{traffic}(n) \quad (3)$$

$$\text{Route}_{TsD} = \sum_{e \in psd} \text{Dist}(e) + \sum_{e \in psd} \text{Tolerance}(n) \quad (4)$$

The network is optimized with the intention of route management by considering energy enhancement in each transmission during the duty cycle. The presence of nodes from Source S at $n \in R$ belongs with intermediate clusters. The dynamic transmission is updated on RT having the energy consumption rate at K number of clusters.

$$Eh = lE_e + l\epsilon_{sd}^2 + (n - 1)l(E_e + E_{BF}) \quad (5)$$

Sensor nodes with low energy consumption and low latency are considered, and non-cluster formation active nodes are avoided during transmission.

$$Enh = lE_e(1 + k) + l\epsilon_{sd}^2 + klE_{BF} \quad (6)$$

The maximum support energy consumed by the cluster head depends on the transmission range and density of the medium. The Lookup finds the minimal congestion to route the transmission.

$$Ed = lE_e + l\epsilon_{ld}^2 + (n/k - 1)l(E_e + E_{BF}) \quad (7)$$

Non-residual energy consumption nodes are discarded from the routing, and relay constraints are considered to add-on nodes for balancing routing in each duty cycle.

$$End = lE_e + l\epsilon_{sd}^2 \quad (8)$$

Initial values are:

$$\epsilon_s = 10 \text{ pJ/bit/m}^2, \quad E_e = 50 \text{ nJ/bit} \quad (9)$$

$$\epsilon_l = 0.001 \text{ pJ/bit/m}^4, \quad E_{BF} = 5 \text{ nJ/bit} \quad (10)$$

The minimal energy consumption actual rate is given by:

$$ETx_{kd} = \begin{cases} E_{elec} \cdot k + \epsilon_{fs} \cdot k \cdot d^2 & \text{if } d < d_0 \\ E_{elec} \cdot k + \epsilon_{mp} \cdot k \cdot d^4 & \text{if } d \geq d_0 \end{cases} \quad (11)$$

$$d_0 = \sqrt{\frac{\epsilon_{fs}}{\epsilon_{mp}}} \quad (12)$$

$$ERx(k) = E_{elec} \cdot k \quad (13)$$

The sensor elective consumption is E_{elec} in lookup transmission on each amplifying unit in a sliding window as reference duty cycle c_{mp} in a multipath cooperative transmission medium. This improves optimized transmission at equalized energy paths with maximum probability to improve the network's lifetime.

B. Probability Traffic Density Approximation Rate

The probability rate finds the maximum support during transmission, which has higher attention to QoS services to deliver data between source and destination without traffic. This considers the minimum residual energy route in the traffic medium from lookup transmission to reduce delay tolerance, with maximum link stability rate under probability density ratio to route the data transmission.

Traffic constraints depend on delay constraints, where the packet transmits by the average mean rate of requesting queuing from one node to another. This creates a support link by choosing the cost D_{cost} to transmit the packet from node J to K without delay. The probability finds the low tolerance delay in route as low traffic delay (DQ) with transmission level (DT) and propagation (DP) level in N number of routes containing 'n' nodes. The low-cost traffic constraints are defined as:

$$D_{cost}(N_j N_k) = DQ_{N_j N_k} + DT_{N_j N_k} + DP_{N_j N_k} \quad (14)$$

$$DQ_{N_j N_k} = \frac{1}{\mu - \lambda_k} \quad (15)$$

The actual traffic is the mean rate from the service on route μ and λ is the new transmission from source node K with a new traffic less propagation link defined as:

$$DT_{N_j N_k} = \frac{I_j}{\Psi} \quad (16)$$

The average consumption without delay tolerance Ψ from the traffic mean is I_j depending on the traffic.

$$DP_{N_j N_k} = \frac{1}{\Psi} \left[1 + \frac{(I_j + I_k)}{\Psi} \right] \quad (17)$$

The physical substantive link without traffic decreases the delay (f^D) from $\|N_j - N_k\|$ by consuming J node without delay, thus increasing the speed of transmission and traffic.

$$f^D(N_j, BS) = \sum_{x \in \{k, U, BS\}} D_{cost}(N_j, N_k) \quad (18)$$

The base station comments on the delay tolerance on each route U as a group of intermediate transmission nodes from one another handover communication J without delay analysis to make an effective link.

C. Link Establishment

This method creates a list of neighbors according to the values in the neighbor table and selects each node with high energy, low traffic, and low moving speed depending on the above factors. Based on this, the method identifies the nodes and establishes connections. The created connection is used for data transfer. This method also uses hop count to establish the list.

D. Q-Learning Based Route Selection

The routing is based on the proposed model of data transmission on different metrics. First, it uses the route discovery process to identify the route. This method calculates Q-Probabilistic Link Availability (Q-PLA) measurement for each identified path and provides QoS-SM (QoS-supported metrics) based on traffic on the route, average neighbors, hop count, and speed of each node on the route. A single method route is selected for data transmission, depending on

Estimate QoS SM =

$$\left(\frac{\sum_{i=1}^{\text{size}(R)} \text{Traffic} \in R(i)}{\text{size}(R)} \right) \left(\frac{\sum_{i=1}^{\text{size}(R)} \text{Flow} \in R(i)}{\text{size}(R)} \right) \left(\frac{\sum_{i=1}^{\text{size}(R)} \text{Speed} \in R(i)}{\text{size}(R)} \right) \times \text{PLA}(R) \quad (19)$$

The working principle of the route selection algorithm is presented as the method described above for probabilistic link availability metric calculation, and the QoS-SM value of the calculation is for each route.

E. Petal Spider Ant Colony Cluster

The ant colony is optimized with decisive layers to form a spider optimization algorithm. The nodes are recognized, and for each of them, a weighted layer is computed for the location proximity value. The location proximity is measured based on the average distance with the other cluster energetic nodes. By selecting the cluster head based on location proximity, any node in the cluster can reach the cluster head with minimal hop count, which in turn reduces latency values. According to the traffic and throughput expectation, each sensor is measured by means of the traffic and throughput expectation for the cluster traffic support (CTS). Finally, a sensor with maximum CTS is selected to execute data transmission. Since each type of traffic has its own requirements for the QoS metric, the goal of routing selection is to select an accessible path (Pt), where f represents each traffic class defined by the application running in the WSN. Each class of traffic connects source CH nodes with different QoS requirements to balance traffic distribution and improve network performance. This can be expressed as the objective function:

$$\begin{aligned} f(P_t) = & \omega_p(1 - \text{packet loss}(P_t)) \\ & + \omega_d(D_{\max} - \text{delay}(P_t)) \\ & + \omega_m(\text{memory}(P_t) - M_{\min}) \\ & + \omega_b(\text{bandwidth}(P_t) - B_{\min}) \\ & + \omega_e(\text{residual energy}(P_t) - E_{\min}) \end{aligned} \quad (20)$$

where $\omega_p, \omega_d, \omega_m, \omega_b, \omega_e$ respectively denote the weights.

IV. SIMULATION RESULT PERFORMANCE

A 100-node distributed over a 900x900m network area is presented in the simulation parameter Table I.

TABLE I
PARAMETERS FOR SIMULATION

Parameter	Values
Simulation tool	NS2
Node placement	Random placement
Number of Nodes	100
Simulation area	300m x 300m
Simulation Time	500sec
Routing Protocol	TCP
Network Type	Static (Fixed Network)
Data Packet Size	512 KB
Channel Type	Wireless

The novel characteristics estimate the routing performance, throughput, packet drop rate, delay, and security performance. In the NS2 environment, nodes are randomly deployed with 20-100 nodes. This section presents the proposed simulation result snapshots, and parameters' performance are discussed in detail. The evaluated parameters are routing performance, security, throughput, packet drop ratio, latency, and energy consumption.

Figure 2, Figure 3, and Table III compare the proposed EOLSRA-CS2RA with existing methods PAR, PSO, and FAPRP. Routing performance during data transfer selects the

TABLE II
IMPACT OF ROUTING PERFORMANCE

Nodes/Methods	PAR	PSO	FAPRP	EOLSRA-CS2RA
25	70	73	78	84
50	76	79	82	87
75	81	87	91	94
100	85	92	94	96

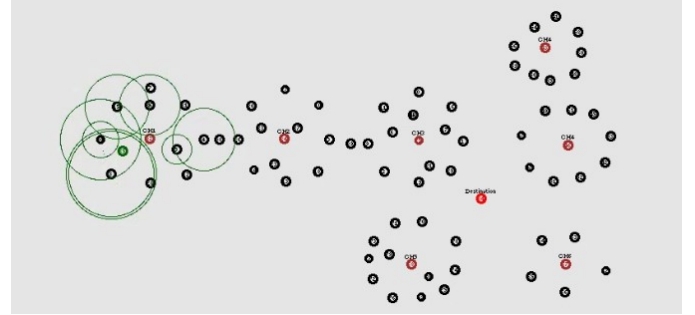


Fig. 2. Routing Performance

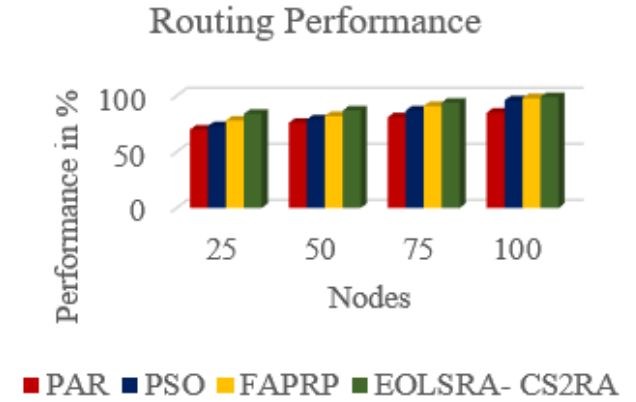


Fig. 3. Impact of Routing Performance

non-malicious nodes for efficient packet communication. The X-axis represents the number of nodes, increasing by 25 per stage. The Y-axis represents the routing performance achieved by each method, along with the number of nodes. The proposed algorithm achieves 84% for 25 nodes, 87% for 50 nodes, 94% for 75 nodes, and 99% for 100 nodes. Thus, the proposed method achieves higher routing performance compared to previous methods.

TABLE III
IMPACT OF THROUGHPUT PERFORMANCE

Nodes/Methods	PAR	PSO	FAPRP	EOLSRA-CS2RA
25	69	74	77	83
50	75	79	84	88
75	79	87	93	95
100	81	84	94	97

Figure 4 analyzes the cluster performance to select the node between the cluster heads and send and receive packets.

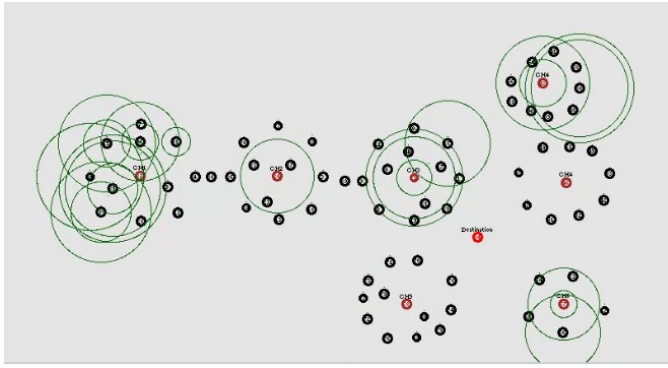


Fig. 4. Screenshot for Cluster Performance

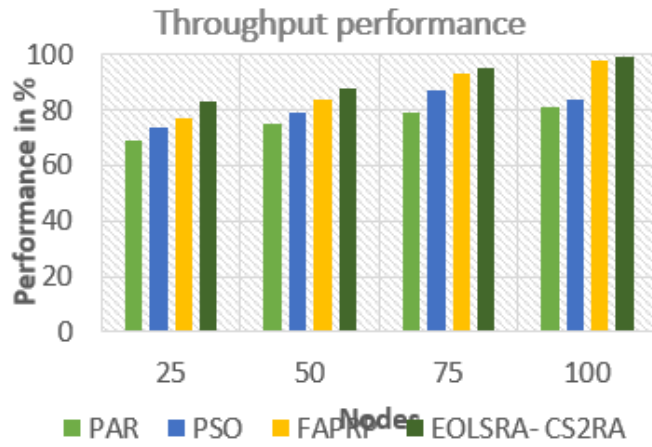


Fig. 5. Impact of Throughput Performance

Figure 5 and Table III compare the proposed EOLSRA-CS2RA with existing methods PAR, PSO, and FAPRP. Throughput performance improves the overall network performance. The X-axis represents the number of nodes, increasing by 25 per stage. The Y-axis represents the total throughput efficiency achieved by each method, along with the number of nodes. The proposed algorithm achieves 83% for 25 nodes, 88% for 50 nodes, 95% for 75 nodes, and 99% for 100 nodes. By detecting the energy state and behavior of the nodes and thereby improving communication by identifying the malicious node and avoiding it, the proposed system improves throughput performance. Thus, the proposed method achieves higher performance compared to previous methods.

Figure 6 depicts the packet drop ratio performance in percentage using different nodes like 25, 50, 75, and 100. The proposed method achieves a packet drop ratio of 15% for 100 nodes. The proposed method produces a low packet drop ratio compared with previous methods. Packet drop ratio is calculated using the following equation:

$$\text{Packet drop ratio} = \frac{\sum(P_s - P_r)}{\text{Total}(P_s)} \times 100 \quad (21)$$

Equation 21 is used to find the packet loss ratio during data

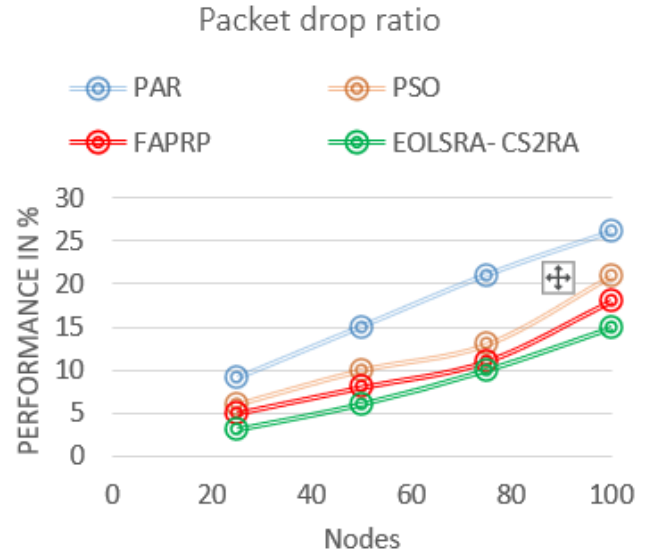


Fig. 6. Impact of Packet Drop Ratio

communication between sender and receiver in the network. Let us assume P_s refers to the packet sent by the sender, P_r refers to the packet received by the receiver.

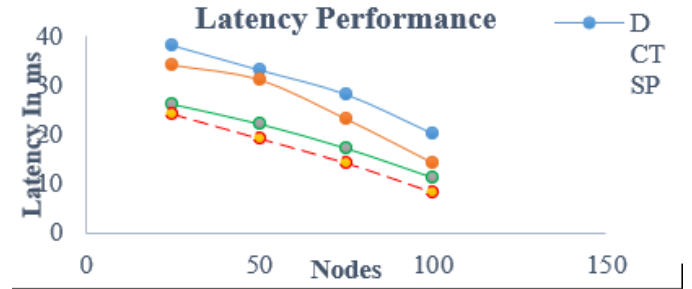


Fig. 7. Impact of Latency Performance

Figure 7 and Table IV compare the proposed EOLSRA-CS2RA with existing PAR, PSO, and FAPRP methods. The X-axis represents the number of nodes, rising by 25 per stage. The Y-axis represents the latency performance of each technique with its nodes. The proposed algorithm achieves 24ms for 25 nodes, 19ms for 50, 14ms for 75, and 8ms for 100 nodes. Here, ms refers to milliseconds. The above figure shows the low latency performance by the proposed algorithm.

TABLE IV
SCHEDULING PERFORMANCE

Nodes/Methods	PAR	PSO	FAPRP	EOLSRA-CS2RA
25	70	78	80	81
50	76	83	86	89
75	82	89	91	94
100	89	91	92	95

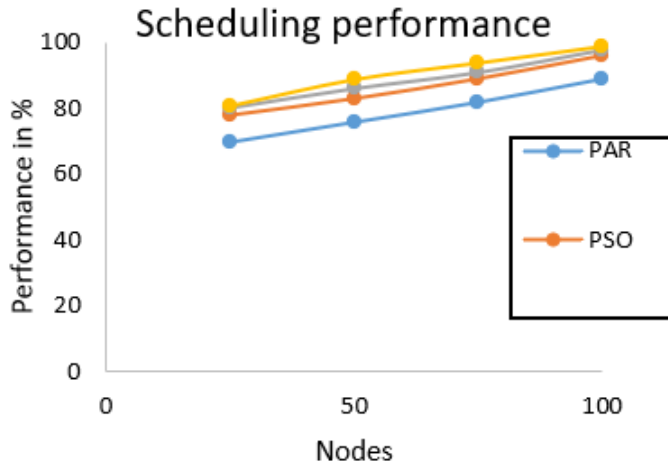


Fig. 8. Impact of Scheduling Performance

Figure 8 and Table IV describe comparison work carried out on the proposed EOLSRA-CS2RA with existing methods PAR, PSO, and FAPRP. Scheduling performance is malicious node detection based on data transfer delay time and node behavior. The X-axis represents the number of nodes, increasing by 25 per stage. The Y-axis represents the security performance achieved by each method, along with the number of nodes. The proposed algorithm achieves 81% for 25 nodes, 89% for 50 nodes, 94% for 75 nodes, and 99% for 100 nodes.

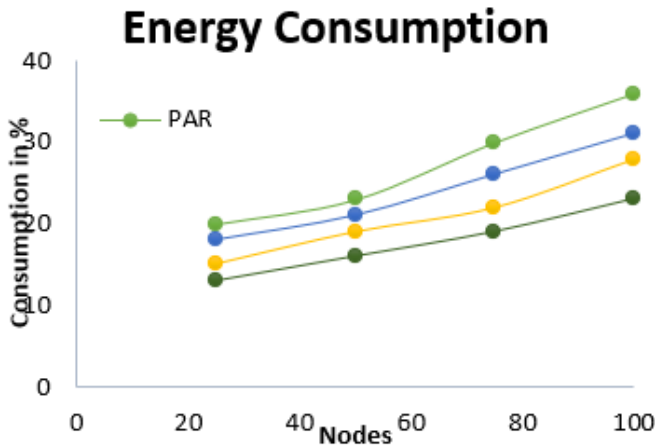


Fig. 9. Impact of Energy Consumption

Figure 9 compares the proposed EOLSRA-CS2RA with existing PAR, PSO, and FAPRP methods. Less power consumption increases the efficiency of the overall network performance. The X-axis represents the number of nodes, rising by 25 per stage. The Y-axis represents the energy consumption performance of each technique with its nodes. The proposed algorithm achieves 13% for 25 nodes, 16% for 50 nodes, 19% for 75 nodes, and 23% for 100 nodes. The above figure shows the low power consumption by the proposed algorithm.

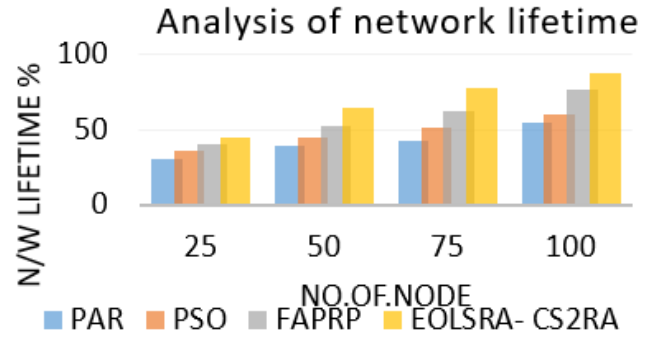


Fig. 10. Analysis of Network Lifetime

Figure 10 shows the Network lifetime Performance analysis based on the packets comparing the existing and proposed system, in the proposed method EOLSRA- CS2RA is 90% accuracy better than previous methods.

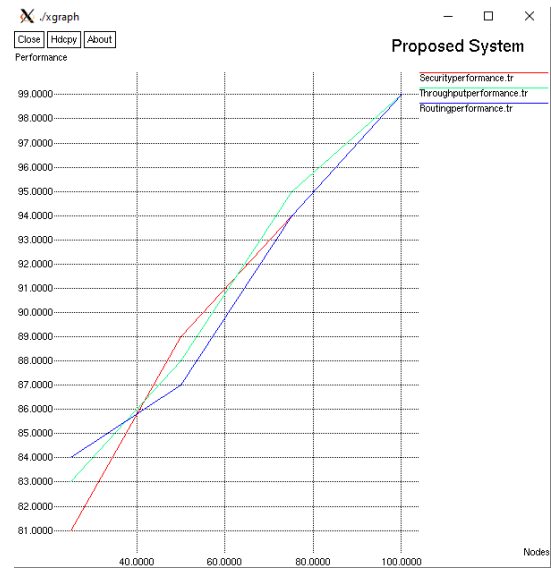


Fig. 11. Overall performance to improve QoS performance

Figure 11 illustrates overall performance simulation screenshots to improve QoS performance implemented on NS2. The X-axis represents the number of nodes, rising to 20 per stage. The Y-axis represents the overall performance of the proposed technique with its nodes.

V. CONCLUSION

This paper presented a Cooperative self-scheduling routing approach (CS2RA) based on Energy Efficient Optimal Link stability routing allocation (EOLSRA), designed for improving the quality of service in Wireless Sensor Networks. The proposed approach achieves the best transmission to improve performance. The dynamic network topology medium is constructed with Neighbor Stable Link Connectivity Based Routing supporting dynamic source routing. With the support of Probability Traffic Density Approximation Rate (PTDAR),

energy stabilization is improved. Additionally, the Petal Spider Ant Colony Cluster (PSACC) is constructed for link stability and route optimization. The proposed protocol improves scheduling and redirects packets during communication. The experimental results show that the throughput is 95.4%, routing performance is 97.1%, energy consumption is 23%, latency is 8ms, and scheduling performance is 95.7% for 100 nodes, compared with previous methods like PAR, PSO, and FAPAR.

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